

PRAVEEN K

SOC Analyst | Junior Threat Intelligence Analyst | Cybersecurity Analyst

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PROFESSIONAL SUMMARY

Entry-level SOC Analyst with hands-on SIEM and threat detection experience, including a self-built home lab simulating real-world attacks across the MITRE ATT&CK kill chain. Engineered Pherion – a Python-based IDS with 37+ detection rules spanning SYN Flood, DNS Tunneling, C2 Beacons, and Lateral Movement – integrated with Splunk for log ingestion and alert correlation. Proficient in IOC analysis, PCAP investigation, alert triage, and false positive reduction. Seeking an L1 SOC Analyst role to contribute to security operations and incident response.

TECHNICAL SKILLS

SIEM & Monitoring : Splunk (log ingestion, alert rules, dashboards, triage), Security Event Monitoring, Log Analysis, Alert Investigation, False Positive Reduction, Incident Ticketing.

Security Tools : Wireshark, Nmap, Burp Suite, Nessus, Metasploit, Kali Linux

Threat Detection & Response : IDS/IPS, IOC/IOA Analysis, Threat Hunting, Threat Intelligence, Malware Analysis, Incident Response, PCAP Analysis, Vulnerability Assessment, RCA (Root Cause Analysis), Alert Escalation, Playbook/Runbook Execution.

Frameworks & Standards : MITRE ATT&CK, Cyber Kill Chain, OWASP Top 10, NIST Cybersecurity Framework (awareness), IAM Fundamentals, CVSS Scoring (awareness).

Networking & Protocols : TCP/IP, DNS, HTTP/HTTPS, VPN, Firewalls, ARP, DHCP, Packet Analysis, Network Scanning, Port Enumeration.

OS & Platforms : Kali Linux, Windows Security, Linux CLI, VirtualBox, Virtual Lab Environments.

Programming & Scripting : Python (security scripting, automation), Bash/Shell Scripting, PowerShell (basics).

Concepts : AES-256 Encryption, Steganography, Zero Trust Architecture, EDR (CrowdStrike/SentinelOne), SOAR.

PROJECTS

Pherion – AI-Powered IDS & SOC Monitoring Platform

Tech Stack: Python, Scapy, Scikit-learn, Pandas, SQLite, JSON, Splunk

- Engineered 37+ detection rules mapped to MITRE ATT&CK TTPs (T1046 – Port Scan, T1071 – C2 Beacons, T1110 – Brute Force, T1557 – ARP Spoofing, T1048 – DNS Tunneling, T1021 – Lateral Movement) for structured threat intelligence reporting.
- Integrated ML-based anomaly detection with auto-tuning thresholds, reducing false positive alert noise by ~30% in lab testing – improving analyst triage efficiency.
- Generated SIEM-ready JSON logs ingested into Splunk, enabling real-time alert investigation and SOC workflow simulation with dashboard visibility.

Praxion – USB Malware Scanner & Threat Detection Tool

Tech Stack: Python, YARA, PE File Analysis, VirusTotal API, ssdeep

- Automated IOC identification using VirusTotal API and ssdeep fuzzy hashing, enabling faster malware triage and threat classification decisions.
- Built YARA rule matching engine with heuristic scanning and PE inspection for detecting known and unknown malware families on USB devices.
- Implemented forensic-grade logging and secure quarantine system, producing structured evidence output suitable for incident documentation.

PRISM – SIEM Rule Generator from Raw Logs

Tech Stack: Python, Rich, Regex, Sigma YAML, Splunk SPL, JSON, MITRE ATT&CK

- Architected a regex parsing pipeline for auth.log, Apache access logs, and Windows Security events – detecting failed logins, port scan surges, and privileged execution attempts per source IP against defined thresholds.
- Engineered automated rule translation layer converting raw log insights into deployable Sigma YAML and Splunk SPL detection rules, with a built-in validation module cross-referencing generated rules against sample malicious logs before deployment.
- Designed a MITRE ATT&CK mapping engine assigning tactics and CVE identifiers to extracted artifacts, with normalized JSON output for universal SIEM ingestion.

HOME LAB – SOC ANALYST SIMULATION ENVIRONMENT

Tech Stack: VirtualBox, Kali Linux, Windows 10, Splunk Enterprise, Wireshark

- Built a segmented dual-VM SOC lab (Kali Linux / Windows 10 + Splunk) to simulate real-world attack scenarios including T1110 (Brute Force), T1078 (Valid Accounts), T1046 (Port Scan), and T1021 (Lateral Movement).
- Performed L1 analyst triage: detected alerts → investigated event logs → correlated IOCs → escalated findings → documented incidents in structured ticketing format (simulating ServiceNow/Jira workflow).
- Created custom Splunk detection rules with false positive tuning; followed NIST IR playbook (Identify → Contain → Eradicate → Recover → Document) for each simulated incident.

INDEPENDENT RESEARCH / CVE ANALYSIS

CVE-2021-44228 | Log4Shell RCE Analysis

- Analyzed Log4Shell RCE flaw in Apache Log4j (v2.0–2.14.1) where malicious JNDI strings in HTTP headers trigger remote code execution; mapped attack chain: Initial Access (T1190) → Code Execution (T1059) → C2 Callback (T1071).
- Built Splunk query to detect JNDI patterns in web logs and alert on unexpected outbound LDAP traffic – simulating SOC L1 triage workflow.
- Documented patch (Log4j 2.15.0), WAF rules, and input sanitization as mitigations in a structured SOC analyst incident brief.

CVE-2026-25769 (Critical RCE) in Wazuh SIEM

- Investigated a critical RCE vulnerability in Wazuh SIEM caused by insecure deserialization in cluster communications – analyzed how a rogue worker node could execute arbitrary code on the master node with root privileges, leading to full system compromise, and mapped the finding to MITRE ATT&CK T1203 (Exploitation for Client Execution).
- Analyzed exploitation impact, detection indicators, and remediation steps including upgrade to the patched Wazuh release – simulating real-world SOC vulnerability triage and patch management workflow.

HANDS-ON TRAINING & INTERNSHIP

Hackup Technology pvt.Ltd – Cybersecurity Internship (1 Month)

- Performed web application security testing using Burp Suite; identified SQLi, XSS, and IDOR vulnerabilities mapped to OWASP Top 10.
- Analyzed network traffic for suspicious packets/IOCs using Wireshark; conducted network scanning and service enumeration with Nmap to identify open ports and attack surfaces.
- Executed controlled exploitation using Metasploit on vulnerable lab machines; documented findings and remediation steps including privilege escalation analysis.

Vital Skills x Techkriti – IIT Kanpur - Ethical Hacking & Network Security Internship (1 Mounth)

- Practiced full penetration testing lifecycle (Recon → Scan → Exploit → Post-Exploit) aligned with Cyber Kill Chain; simulated MITM, brute-force, and network sniffing attacks to understand attacker TTPs and defensive controls.
- Performed vulnerability assessments using Nmap and Burp Suite; analyzed PCAP files in Wireshark to trace attack paths and produced risk-rated reports with remediation notes.

EDUCATION

B.E. Computer Science and Engineering

St.Michael College of Engineering and Technology - Kalayarkoil

2022- 2026

CERTIFICATIONS

- Cisco Networking Academy – Ethical Hacker
- Cisco Networking Academy – Network Defence
- Fortinet Training Institute – NSE 3 – Associate in Cybersecurity
- Infosys Springboard – IoT & Electronics

SOFT SKILLS

- Analytical Thinking • Threat Pattern Recognition • Methodical Problem-Solving • Attention to Detail • Team Collaboration • Technical Documentation & Reporting
- Adaptability Under Pressure • Continuous Learning & Upskilling